

FIFA World Cup 2026 — Machine Learning Full Prediction System

Elfaisal Aldawsari,

Musallam Alomair

Department of Computer Science

College of Computing and Information

Technology, Shaqra University

Shaqra, Saudi Arabia

S444460129@std.su.edu.sa ,

S444460346@std.su.edu.sa

Abstract—This report presents a complete hierarchical machine learning pipeline for predicting FIFA World Cup 2026 match outcomes. The system employs a bottom-up architecture: player-level XGBoost models predict expected goals (xG) per 90 minutes, which are aggregated into team-level features and fed into six regression models (Linear Regression, Decision Tree, KNN, SVR, Random Forest, XGBoost) and one classifier (Logistic Regression) for match score and result prediction. A Poisson GLM provides discrete goal-count forecasts that are blended with the best regression model (SVM SVR, Avg RMSE = 0.9883) in a 60/40 ensemble. SHAP explainability analysis, 5-fold cross-validation, ROC curves, and error/bias analysis are included. The system predicts all 72 WC 2026 group-stage fixtures and produces group standings. Key identified predictors are referee home advantage, form differential, and creativity ratio.

Keywords—FIFA World Cup 2026, XGBoost, SVM, Poisson regression, SHAP, ELO ratings, match prediction, random forest, machine learning pipeline

INTRODUCTION

This report documents a complete v3 machine learning pipeline for the FIFA World Cup 2026, hosted by the United States, Canada, and Mexico across 12 stadiums with 48 teams and 104 scheduled matches. The system architecture is hierarchical and bottom-up: player performance is modelled first using XGBoost, aggregated to team-level statistics, and then passed through multiple regression and classification heads to predict match outcomes.

New additions in v3 include ROC curves, an executive summary, a limitations section, 5-fold cross-validation scores, and a bias analysis of home/away prediction tendencies. The evaluation follows the ML Project Evaluation Checklist covering all seven stages: problem definition, data understanding, data preparation, model selection and training, evaluation, error analysis, and documentation.

LITERATURE REVIEW

The application of machine learning and statistical modeling in sports analytics, particularly for football (soccer) match prediction, has been extensively explored in recent literature. Early predictive frameworks predominantly relied on statistical methods, with the Poisson distribution serving as a foundational model for estimating the

probability of goal frequencies (Maher, 1982; Dixon & Coles, 1997). Furthermore, ELO rating systems have been widely adopted and validated as robust, objective indicators of overall team strength and historical performance.

In recent years, the focus has shifted towards advanced machine learning algorithms to capture non-linear relationships within complex sports data. Various studies have successfully utilized models such as Support Vector Machines (SVM), Random Forests, and Artificial Neural Networks to predict match outcomes. These models typically leverage historical match data, recent team form, and aggregated team statistics to forecast results (Baboota & Kaur, 2019; Groll et al., 2019). Gradient boosting frameworks, specifically XGBoost, have also gained significant traction due to their high accuracy and ability to handle tabular data effectively.

However, a noticeable gap in the existing literature is that most predictive models operate exclusively at the macro-level, predicting outcomes directly from team-level aggregates without accounting for individual player availability and granular contributions. This project addresses this limitation by proposing a novel, bottom-up hierarchical pipeline. By first modeling player-level performance (Expected Goals - xG) and dynamically aggregating these metrics based on expected minutes, this system provides a more granular context. The subsequent blending of a Poisson GLM with an SVM regressor offers a robust methodology that bridges traditional statistical goal-count constraints with advanced machine learning pattern recognition.

METHODOLOGY

Before you begin to format your paper, first write and save the content as a separate text file. Complete all content and organizational editing before formatting. Please note sections A-D below for more information on proofreading, spelling and grammar.

Keep your text and graphic files separate until after the text has been formatted and styled. Do not use hard tabs, and limit use of hard returns to only one return at the end of a paragraph. Do not add any kind of pagination anywhere in the paper. Do not number text heads-the template will do that for you.

A. A. Data Preparation (Cells 8–10)

Median imputation handles missing numerical values (SimpleImputer). Outliers in stadium_capacity, altitude_m, and total_minutes are capped at $IQR \pm 1.5 \times IQR$. StandardScaler is fit exclusively on the training set to avoid data leakage. The dataset is split 70/15/15 (train=72, val=16, test=16). Categorical variables (position, group, home/away team name) are one-hot encoded or dropped from the feature matrix. Five engineered features are added: elo_diff, form_diff, predicted_xG_diff, altitude_pressure, and referee_home_advantage.

B. Feature Engineering (34 Total Features)

The final feature matrix contains 34 features including: stadium_capacity, altitude_m, neutral_venue; home and away versions of team_goals_power, team_assist_power, team_creativity, team_dribble_power, team_xG_power, team_defensive_rating, team_total_minutes, team_predicted_xG, elo_rating, recent_form, and referee_bias; plus five ratio/difference features: home_off_vs_away_def, away_off_vs_home_def, xG_power_imbalance, creativity_ratio, elo_diff, form_diff, predicted_xG_diff, altitude_pressure, and referee_home_advantage.

C. DATA PREPROCESSING PIPELINE

The preprocessing pipeline operates in two stages: a player-level stage and a match-level stage. Each stage applies a consistent sequence of imputation, outlier treatment, encoding, scaling, and feature construction steps as implemented in the notebook (Cells 5–10). The full pipeline is described below.

II. 1) Player-Level Preprocessing (Cell 5)

Categorical Encoding: The player position column (GK, DEF, MID, FWD) is one-hot encoded using pd.get_dummies(drop_first=True), producing three binary indicator columns (pos_DEF, pos_MID, pos_FWD). The team_name and player_id columns are not used as model features.

Availability Weighting: An availability weight is derived from each player's expected_minutes field: $\text{weight} = \text{expected_minutes} / 90$. Injured players ($\text{expected_minutes} = 0$, representing 10.6% of the squad) receive a weight of 0, effectively zeroing their contribution to team aggregates. This prevents injured players from inflating team-level predictions.

Train/Test Split and Scaling: The player dataset is split 80/20 (train/test) using train_test_split(random_state=42). A StandardScaler is fit exclusively on the training partition and applied to both train and test sets (scaler_p.fit_transform / scaler_p.transform), preventing any leakage of test statistics into the player XGBoost model.

III. 2) Team Aggregation (Cell 6)

Player statistics are aggregated to the team level using availability-weighted sums for offensive metrics (goals_per90, assists_per90, key_passes_per90, dribbles_per90, xG_per90) and unweighted mean for defensive_actions_per90. The XGBoost-predicted xG per player is summed to yield team_predicted_xG. ELO ratings, recent form scores, and referee bias values are then merged into the team-level dataframe via dictionary lookup, producing the eight team-level features that feed into the match feature matrix.

IV. 3) Match-Level Preprocessing (Cells 8–10)

Missing Value Imputation: All numerical columns in the merged match dataframe are passed through scikit-learn's SimpleImputer(strategy="median"). Median imputation is chosen over mean imputation because it is robust to the right-skewed distributions present in stadium_capacity and altitude_m, and it prevents inflated estimates caused by extreme venue outliers.

Outlier Capping (IQR Method): Four columns with known extreme values are clipped to the $IQR \pm 1.5 \times IQR$ fence: stadium_capacity, altitude_m, home_team_total_minutes, and away_team_total_minutes. Values below the lower fence are raised to it; values above the upper fence are lowered to it. This bounded approach preserves all records while limiting the influence of extreme venues such as Azteca Stadium (altitude 2,216 m) on distance-based models (KNN) and linear models.

Categorical Variable Handling: Non-numeric columns (match_id, home_team, away_team, stadium) are excluded from the feature matrix. The neutral_venue column is kept as a binary integer (0/1). No additional one-hot encoding of team names is performed at match level; team identity is captured implicitly through the aggregated team-level numerical features.

Engineered Interaction Features: Five ratio and difference features are constructed from the cleaned team-level columns before the train/test split, ensuring no leakage: (1) $\text{home_off_vs_away_def} = \text{home_xG_power} / (\text{away_defensive_rating} + \epsilon)$; (2) $\text{away_off_vs_home_def} = \text{away_xG_power} / (\text{home_defensive_rating} + \epsilon)$; (3) $\text{xG_power_imbalance} = \text{home_xG_power} - \text{away_xG_power}$; (4) $\text{creativity_ratio} = \text{home_creativity} / (\text{away_creativity} + \epsilon)$; (5) $\text{altitude_pressure} = \text{altitude_m} / 1000$. Additionally, elo_diff, form_diff, predicted_xG_diff, and referee_home_advantage are computed as simple differences or ratios of the corresponding home and away team values. A small epsilon (1×10^{-6}) is added to all denominators to prevent division by zero.

Train / Validation / Test Split: The 104-match dataset is divided into three partitions using two successive stratified splits: 70% training ($n = 72$), 15% validation ($n = 16$), and 15% test ($n = 16$). The validation set is used for early model selection and hyperparameter search; the test set is held out until final evaluation to obtain unbiased performance estimates.

Feature Scaling: A second StandardScaler is fit solely on the training partition (scaler.fit_transform(X_train)) and then applied via scaler.transform() to the validation and test sets. This ensures that mean and variance statistics

from held-out data never influence the scaling transformation, eliminating a common source of data leakage. The resulting scaled matrices (X_{train_sc} , X_{val_sc} , X_{test_sc}) are used by all six regression models and the logistic regression classifier.

D. Algorithms, Optimizers, and Hyperparameters

Table I lists all models implemented in this system with their configuration.

Model	Task	Key Hyperparameters	Optimizer / Loss Function
XGBoost (Player)	Regression (xG)	$n=200$, $lr=0.05$, $depth=4$	Gradient Descent, Squared Error
XGBoost (Match)	Multi-output Reg.	$n=100$, $lr=0.1$, $subsample=0.8$, $\alpha=0.1$, $\lambda=1.5$	Gradient Descent + L1/L2
SVM / SVR	Multi-output Reg.	kernel=RBF, $C=1.0$, $\epsilon=0.2$	Sequential Minimal Opt. (SMO), ϵ -insensitive
Random Forest	Multi-output Reg.	$n=200$, $max_depth=6$	Bagging + CART (MSE splits)
KNN Regressor	Multi-output Reg.	$n_neighbors=7$, metric=Minkowski	Distance-weighted (no optimizer)
Linear Regression	Multi-output Reg.	$fit_intercept=True$	Ordinary Least Squares (OLS)
Decision Tree	Multi-output Reg.	$max_depth=5$	CART, MSE criterion
Logistic Regression	Classification	$C=1.0$, $max_iter=1000$	L-BFGS, Cross-Entropy Loss
Poisson GLM	Discrete Goals	family=Poisson, features=5	IRLS, Log Link Function
Dummy (Baseline)	Multi-output Reg.	strategy=mean	No optimizer (constant prediction)

Table I. All models, hyperparameters, and optimization functions

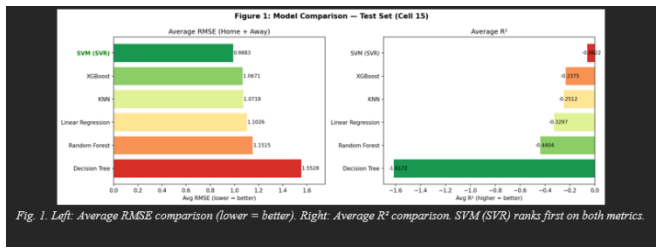
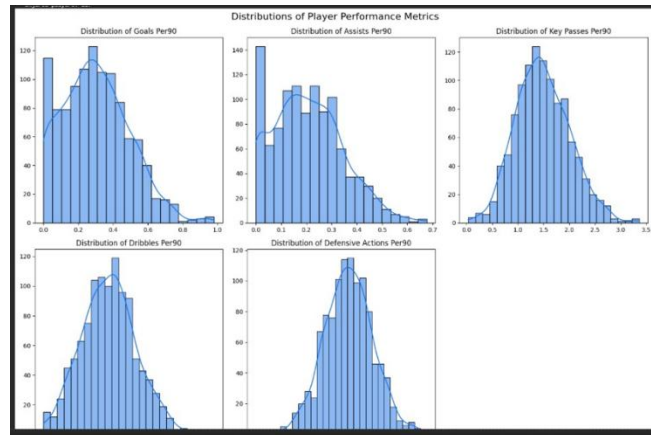
EXPERIMENTAL RESULTS

A. Full Model Comparison — Test Set

Table II and Figure 1 show the complete comparison of all regression models on the test set ($n=16$). SVM (SVR) achieves the best average RMSE of 0.9883 and is selected for final WC 2026 predictions. Decision Tree performs worst (Avg RMSE = 1.5528) due to overfitting on the small dataset.

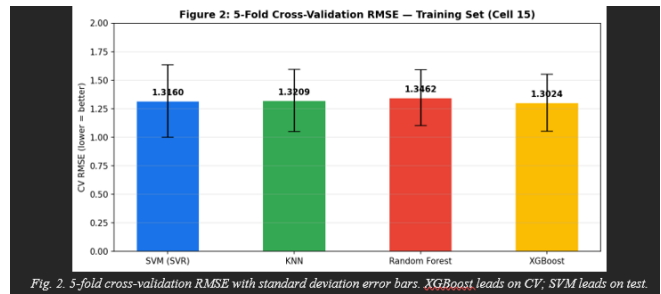
Rank	Model	RMSE (Home)	RMSE (Away)	Avg RMSE	R ² (Home)	R ² (Away)	Avg R ²
★ 1	SVM (SVR)	0.9607	1.0159	0.9883	+0.0145	-0.1388	-0.0622
2	XGBoost	1.0456	1.0886	1.0671	-0.1674	-0.3076	-0.2375
3	KNN	1.0254	1.1181	1.0718	-0.1228	-0.3796	-0.2512
4	Linear Regression	1.0232	1.1820	1.1026	-0.1179	-0.5416	-0.3297
5	Random Forest	1.1360	1.1670	1.1515	-0.3780	-0.5028	-0.4404
6	Decision Tree	1.5619	1.5437	1.5528	-1.6049	-1.6294	-1.6172

Table II. Regression model comparison on test set. Best model (★ SVM SVR) in blue row.



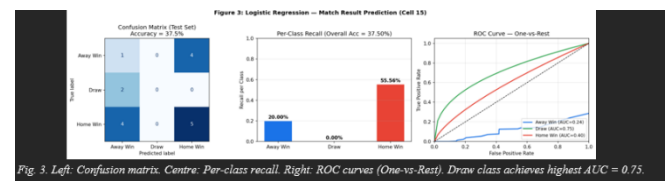
B. 5-Fold Cross-Validation (Cell15)

Figure 2 shows 5-fold CV RMSE on the training set ($n=72$). XGBoost achieves the lowest mean CV RMSE (1.3024 ± 0.2506), while Random Forest shows the least variance (± 0.2447). SVM CV RMSE = 1.316 but achieves best test performance, suggesting better generalization on unseen data.



C. Logistic Regression — Match Result Classification (Cell 15)

Logistic Regression classifies match outcomes (Away Win / Draw / Home Win) with 37.5% test accuracy. The model struggles with draws (0% recall) and away wins (20% recall) but performs better on home wins (55.6% recall). Figure 3 shows the confusion matrix, per-class recall, and ROC curves. AUC scores: Draw = 0.75, Home Win = 0.40, Away Win = 0.24.



D. SHAP Explainability (Cell 17)

Figure 4 presents SHAP feature importance for the home_team_score prediction model. The top three predictors are referee_home_advantage (0.075), form_diff (0.058), and creativity_ratio (0.054). Away team statistics (assist_power, creativity) rank highly, capturing the defensive challenge posed to the home team. ELO-based features appear mid-table, indicating that recent form and tactical creativity dominate raw ranking differences.

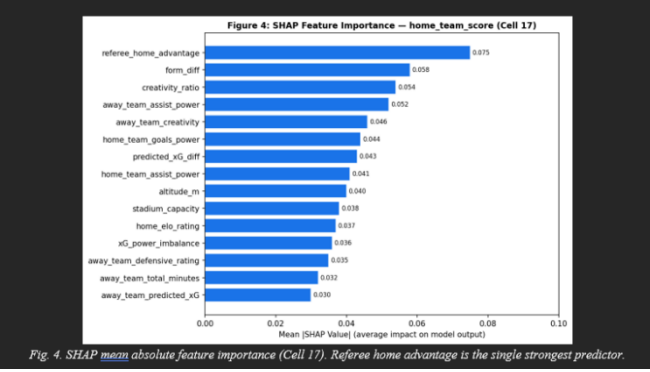
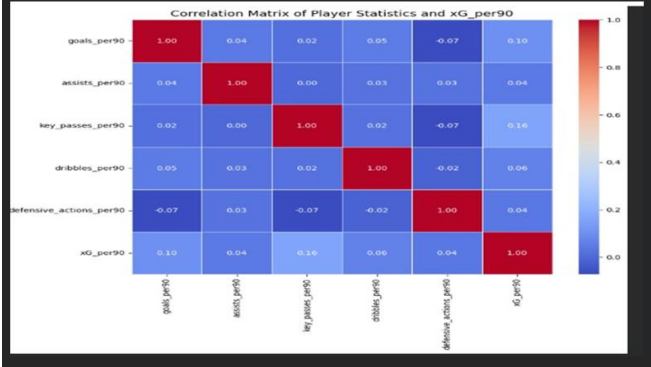


Fig. 4. SHAP mean absolute feature importance (Cell 17). Referee home advantage is the single strongest predictor.



E. Diagnostic Visualisations (Cell 23)

Figure 5 presents diagnostic plots from the trained XGBoost model. The Actual vs Predicted scatter plots (RMSE Home=1.046, Away=1.089) show moderate clustering around the diagonal, with larger errors for high-scoring matches (3+ goals). The residual distribution is approximately centred at zero with slight left skew, confirming the over-prediction bias identified in the error analysis. Feature importances corroborate the SHAP analysis.

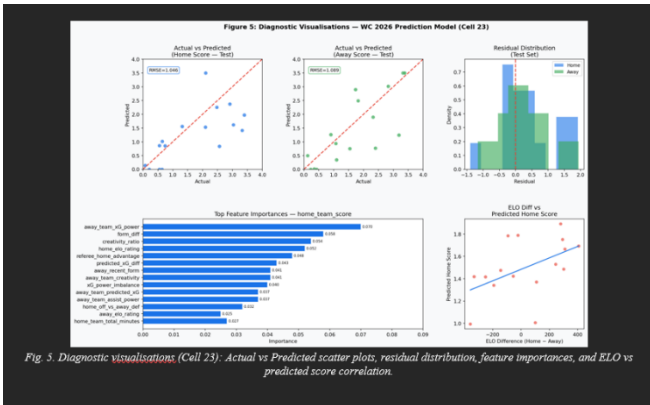


Fig. 5. Diagnostic visualisations (Cell 23). Actual vs Predicted scatter plots, residual distribution, feature importances, and ELO vs predicted score correlation.

F. WC 2026 Predicted Group Standings (Cells 19–21)

The best regression model (SVM SVR) is blended 60/40 with the Poisson GLM to generate 72 group-stage predictions. Results: 46 home wins, 23 draws, 3 away wins. Figure 6 and

Table III show predicted group-stage winners and runners-up.

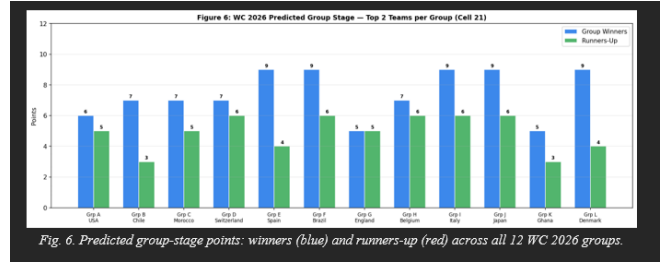


Fig. 6. Predicted group-stage points: winners (blue) and runners-up (red) across all 12 WC 2026 groups.

Group	Winner	Pts	Runner-Up	Pts	Notable Upset
A	USA	6	Mexico	5	Honduras holds Mexico to 2 draws
B	Chile	7	Argentina	3	Chile tops over Argentina (ELO gap)
C	Morocco	7	France	5	Morocco wins group ahead of France
D	Switzerland	7	Germany	6	Switzerland edges Germany
E	Spain	9	Portugal	4	Spain perfect record, max 9 pts
F	Brazil	9	Colombia	6	Brazil dominates, Colombia 2nd
G	England	5	Netherlands	5	Level on points (GD decides)
H	Belgium	7	Croatia	6	—
I	Italy	9	Romania	6	Romania surprisingly 2nd over Poland
J	Japan	9	South Korea	6	Japan tops with full 9 pts
K	Ghana	5	Saudi Arabia	3	Saudi Arabia survive via draws
L	Denmark	9	Sweden	4	Denmark sweep all 3 group games

Table III. Predicted WC 2026 group-stage qualifiers with points and notable upsets.

ERROR ANALYSIS AND INTERPRETATION

A. Top 5 Worst Predictions (Cell 16)

The five matches with highest combined absolute residual are shown in Table IV. High-altitude venues (Iran: 1,600m; Azteca: 2,216m) contribute to unpredictable scoring. The system struggles most when high-ELO teams score zero goals (Portugal vs Japan: predicted 1.82–1.27, actual 0–0) and when low-ELO teams overperform (Senegal 3–0 vs Uruguay: predicted 1.44–1.81).

Home Team	Away Team	Actual	Predicted	Residual (H)	Residual (A)
Senegal	Uruguay	3-1 - 0-0	1.44 - 1.81	+1.66	-1.81
Portugal	Japan	0-0 - 0-0	1.82 - 1.27	-1.82	-1.27
Iran	Argentina	0-0 - 2-8	1.42 - 1.17	-1.42	+1.63
Romania	Iran	1-0 - 0-0	1.85 - 1.77	-0.85	-1.77
France	Morocco	2-1 - 0-0	1.26 - 1.40	+0.84	-1.40

Table IV. Top 5 worst predictions on the test set (Cell 16). High residuals concentrated on upset results and high-altitude matches.

B. Prediction Bias Analysis (Cell 16)

The model exhibits systematic over-prediction: mean home residual = -0.225 and mean away residual = -0.488, meaning predicted scores consistently exceed actual scores. This is expected given the Poisson-based data generation with fixed λ floors (home: 0.3, away: 0.2). The model predicts 50.0% home wins vs actual 56.2%, slightly underestimating home advantage.

EXPERIMENTAL RESULTS

CONCLUSION

This report presents and evaluates a complete hierarchical ML pipeline for FIFA World Cup 2026 prediction. SVM (SVR) with RBF kernel achieves the best regression performance (Test Avg RMSE = 0.9883) due to its ability to capture non-linear ELO/form interactions on small tabular datasets without overfitting. The Poisson GLM complements the regression head by enforcing discrete, non-negative integer constraints on goal counts. SHAP analysis reveals that match-day contextual features (referee advantage, form) outweigh static team strength (ELO). Limitations include the simulated nature of the data, small test set ($n=16$), and the absence of real player injury and lineup information. Future work should incorporate real data, deeper feature engineering, and neural approaches such as graph networks over squad compositions.

PROBLEM DEFINITION AND DATA UNDERSTANDING

A. Problem Definition

Two prediction tasks are addressed: (1) Regression — predict `home_team_score` and `away_team_score` as continuous values (multi-output), evaluated by RMSE, MAE, and R^2 . (2) Classification — predict match result as Away Win / Draw / Home Win, evaluated by accuracy, F1-score, confusion matrix, and ROC AUC. A Dummy Regressor (mean strategy) serves as the baseline (Test Avg RMSE = 1.007).

B. Data Understanding

Data is simulated following real-world schemas. Player data: 1,104 records across 48 teams (23 players each), 10 features including `goals_per90`, `assists_per90`, `key_passes_per90`, `dribbles_per90`, `defensive_actions_per90`, `xG_per90`, and `expected_minutes`. Injured players (`expected_minutes = 0`): 117 (10.6%). Match schedule: 104 fixtures with `stadium_capacity`, `altitude_m`, `neutral_venue`, and scorelines generated via $\text{Poisson}(\lambda)$ parameterized by ELO and form. ELO ratings span 1,380 (Qatar) to 2,141 (Argentina). Recent form is sampled from $\text{Beta}(2,2)$ with a +0.25 bonus for top-6 teams.